

Telecom Customer Attrition Prediction & Retention Analytics

An In-Depth Econometric & Machine Learning Study | Author: Abhiram

1. Executive Summary & Financial Unit Economics

In subscription telecommunications, Net Revenue Retention (NRR) is the primary driver of corporate valuation. Empirical benchmarks indicate that acquiring a new telecom subscriber costs **5× to 7× more** (\$300–\$450 CAC) than proactive retention (\$30–\$60 CRC). A California regional telecommunications provider (*Verizonet*) observed that only **43% of newly acquired subscribers** remained active past their first quarter, driving an annualized churn rate of **26.54%** (1,869 departed accounts out of 7,043 total). This report delivers an end-to-end Machine Learning solution that identifies churners with **94.39% accuracy** and an **ROC-AUC of 0.983**, unlocking an estimated **\$960,900 in annual net revenue savings**.

Metric	Observed Value	Business Context
Total Subscribers Analyzed	7,043 accounts	California service area across 1,000+ zip codes
Baseline Churn Rate	26.54% (1,869 lost)	Significantly higher than industry target (<15%)
Average Monthly Bill (Churned)	\$80.24 / month	Higher price sensitivity among top-tier plan holders
Average Monthly Bill (Retained)	\$61.26 / month	Standard usage accounts exhibit higher retention
Annual Gross Lost Revenue	\$1,457,750 USD	Annualized recurring revenue lost to competitor departures
Champion Model Performance	94.39% Test Acc 0.983 AUC	Gradient Boosting with 5-fold stratified validation
Net Annual Financial Gain	\$960,900 / year	940% ROI over targeted intervention costs (\$102K)

2. Key Diagnostic & Behavioral Discoveries

- **Support Ticket Velocity:** Subscribers logging 0–1 customer service tickets maintain an attrition rate <14%. Upon logging the 2nd support ticket, churn likelihood surges past **75.4%**, and accounts with ≥3 tickets exhibit an **88.9%** churn rate.
- **Contract Volatility:** Month-to-Month contracts represent **88.3% of total churn volume**. In contrast, 1-Year and 2-Year contract holders exhibit churn rates of only 7.8% and 2.3% respectively.
- **Value-Added Services:** Fiber Optic and Cable users without Online Security or Premium Tech Support churn at 3× the rate of accounts with active protection bundles (41.7% vs 14.6%).
- **Tenure Vulnerability:** Subscribers in their first 12 months account for over **52% of all departures**, confirming that early onboarding and first-year resolution dictate lifetime retention.

3. Target Leakage Rectification & Preprocessing Architecture

Target Leakage Resolution: Columns such as *Churn Category* and *Churn Reason* are recorded only after an account is terminated. Including these diagnostic notes during training introduces direct target leakage, causing false 100% training accuracy that collapses on active subscribers. These fields and unique identifiers (*Customer ID*, *City*, *Zip Code*) were strictly excluded from the predictive feature matrix X .

Scikit-Learn ColumnTransformer Pipeline:

- **Numerical Features (15):** Imputed with `SimpleImputer(strategy='median')` and scaled via `StandardScaler()` to standard normal distribution: $z = (x - \mu) / \sigma$.

- **Categorical Features (22):** Imputed with 'Missing' and encoded via OneHotEncoder(drop='first', handle_unknown='ignore') to prevent the dummy variable trap while supporting production inference.

4. Supervised Machine Learning Benchmark (10-Fold Stratified CV)

Five distinct classification algorithms were benchmarked across 10 stratified folds on the 80% training set (5,634 accounts) and evaluated on the 20% holdout test set (1,409 accounts):

Algorithm	10-Fold CV Acc	10-Fold CV F1	Test Acc	Test Precision	Test Recall	Test F1	Test ROC-AUC
Decision Tree (Depth 6)	88.25%	0.751	87.86%	0.780	0.757	0.768	0.912
Random Forest (200 Trees)	92.01%	0.834	92.55%	0.927	0.781	0.848	0.968
Support Vector Machine (SVM)	92.83%	0.858	93.75%	0.901	0.858	0.879	0.975
Logistic Regression (L2)	92.92%	0.862	93.83%	0.902	0.861	0.881	0.976
Gradient Boosting (Champion)	93.43%	0.868	94.39%	0.928	0.856	0.890	0.983

5. Champion Model Diagnostics (Gradient Boosting)

Holdout Confusion Matrix (N = 1,409 accounts):

- **True Negatives (TN = 1,010):** 97.6% of retained subscribers correctly classified.
- **True Positives (TP = 320):** 85.6% of all actual churners successfully intercepted prior to departure.
- **False Positives (FP = 25):** Extremely low false alarm rate (2.4%), preventing wasted promotional credits.
- **False Negatives (FN = 54):** Only 3.8% of test cohort departed without model detection.

Top Predictive Drivers (MDI Feature Importance):

1. **Total Customer Service Requests (28.4%):** Leading indicator of subscriber friction.
2. **Customer Satisfaction Rate (16.2%):** Direct quantified sentiment score.
3. **Contract Type (Month-to-Month) (12.5%):** Lowest barrier to cancellation.
4. **Monthly Charge (8.7%):** Price sensitivity inflection point.
5. **Product/Service Issues Reported (6.1%):** Unresolved technical hardware complaints.
6. **Tenure in Months (5.3%):** Account maturity and loyalty curve.

6. Strategic 4-Pillar Retention Playbook & ROI

Pillar	Target Segment	Actionable Business Intervention	Expected Impact
P1: Support SLA Escalation	Accounts with ≥ 2 support tickets	Automate VIP routing to senior account managers; provide a \$15 bill credit upon 2nd resolution.	Reduces ticket churn from 75% to <30%
P2: Contract Migration	Month-to-Month (Tenure < 12m)	Deploy personalized 15% rate-lock discount on 1-Year or 2-Year plan commitments.	Locks in recurring revenue; churn drops 75%
P3: Hardware & Network QA	Fiber/Cable users with high bills	Complimentary Wi-Fi 6 router upgrades and 60 days of free Premium Tech Support.	Eliminates competitor speed departures
P4: Viral Referral Rewards	Single-line accounts (0 referrals)	Launch 'Give \$25, Get \$25' referral credits for both referrer and referee.	Builds sticky household network effects

7. Technical Conclusion & System Deployment

By enforcing strict leakage boundaries, deploying a robust Scikit-Learn preprocessing pipeline, and implementing the **Gradient Boosting Model** (Holdout Accuracy: **94.39%**, ROC-AUC: **0.983**), telecommunications executives gain a high-precision retention engine capable of saving over **\$960,000 annually** in recurring revenue. All underlying code and executed notebooks are available in the repository.